Year 13 Maths - Differential equations

Similar questions - Forming and solving separable equations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A biologist models the number of bacteria N (in thousands) in a culture at time t hours. When $t = 0$ there are 100 thousand, and when $t = 1$ there are 110 thousand.

(a) In the first model, the rate of increase of N is inversely proportional to N. Write down a differential equation for this model and solve it to find N in terms of t. **[5]**

(b) In a second model, $\dfrac{\mathrm{d}P}{\mathrm{d}t} = \dfrac{P^2\mathrm{e}^{-t/2}}{2000}$ where P is the number of bacteria (in thousands) and $P = 500$ when $t = 0$. Solve this equation to show that $P = \dfrac{1000}{1+\mathrm{e}^{-t/2}}$. **[6]**

(c) Compare the long-term behaviour of N and P. **[2]**

2 A curve C has the property that, at every point $P(x,y)$, the gradient of the tangent is twice the gradient of the line joining $(1,3)$ to P.

(a) Express this property as a differential equation. **[2]**

(b) Given that C passes through $(3,11)$ and that $x > 1$, $y > 3$ at all points on C, find the equation of C in the form $y = \mathrm{f}(x)$. **[5]**

(c) Describe fully the sequence of transformations that maps the curve $y = x^2$ onto C. **[3]**

Question 1

(a) $\dfrac{\mathrm{d}N}{\mathrm{d}t} = \dfrac{k}{N}$. $\int N\,\mathrm{d}N = \int k\,\mathrm{d}t$ gives $\frac{1}{2}N^2 = kt + c$. $t = 0$: $c = 5000$. $t = 1$: $6050 = k + 5000$ so $k = 1050$. $N^2 = 10000 + 2100t$, $N = \sqrt{10000 + 2100t}$

(b) $\int P^{-2}\mathrm{d}P = \int \frac{\mathrm{e}^{-t/2}}{2000}\mathrm{d}t$ gives $-\dfrac{1}{P} = -\dfrac{\mathrm{e}^{-t/2}}{1000} + c$. $P = 500$ at $t = 0$: $-\frac{1}{500} = -\frac{1}{1000} + c$, so $c = -\frac{1}{1000}$. Then $\dfrac{1}{P} = \dfrac{\mathrm{e}^{-t/2}+1}{1000}$ and $P = \dfrac{1000}{1+\mathrm{e}^{-t/2}}$

(c) N grows without limit (like $\sqrt{t}$); P tends to the limiting value 1000 thousand as $t \to \infty$

Question 2

(a) $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2(y-3)}{x-1}$

(b) $\int \dfrac{\mathrm{d}y}{y-3} = \int \dfrac{2}{x-1}\mathrm{d}x$: $\ln(y-3) = 2\ln(x-1) + c$, so $y - 3 = A(x-1)^2$. At $(3,11)$: $8 = 4A$, $A = 2$. $y = 3 + 2(x-1)^2$

(c) Translation by $\begin{pmatrix}1\\0\end{pmatrix}$, then a stretch parallel to the y-axis with scale factor 2, then a translation by $\begin{pmatrix}0\\3\end{pmatrix}$